

one solution that does offer to work with most leading HMDs. But it has its own drawbacks which we have discussed earlier. Mainly, the Virtuix omni doesn't offer as much freedom as you would expect at that price point. Users are still constrained by a harness which doesn't allow them to duck/crouch when they want to.



The Omni does fall short of close-to-natural locomotion in VR.

A possibly better alternative was demonstrated at CES 2016, called the Infinadeck. Although it also calls itself an omnidirectional device, it is closer to a treadmill than Virtuix's static floor approach. At approximately 5 feet by 5 feet, the platform allows you to walk in any direction infinitely as the floor moves under you at a maximum speed of 10kmph, which is an average jogging speed. George Burger, inventor of Infinadeck explains that the entire setup consists of only one belt and two motors. This design provides two main benefits, - two motors instead of multiple cuts the cost largely, and a single belt removes the necessity to synchronize the RPM of the motors across belts which would have been necessary to walk safely on this platform. The treadmill comes with a safety harness to prevent users who walk too fast from falling off and also allows users to crouch or duck as necessary. Over the years, the machine has improved its response